

Research article

Layers of injustice: A distributional assessment of toxic chemical facilities, releases, and cleanups

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ABSTRACT

The Resource Conservation and Recovery Act of 1976 (RCRA) is a cornerstone of environmental policy in the United States. The law regulates the generation, transportation, storage, and disposal of hazardous chemicals. Although regulated, hazardous releases can still occur at RCRA sites due to flawed equipment, human error, and dated historical practices. Releases are investigated and remediated through what is known as a Corrective Action (CA). Using Census data and a novel dataset of RCRA facilities across the contiguous U.S., we examine the possibility of systematic inequities with regards to the (i) siting of RCRA facilities, (ii) occurrence of releases and CAs, (iii) duration of CAs, and (iv) permanence of remediation methods. We find evidence of disproportionate impacts across racial, ethnic, and income dimensions. The results vary, however, depending on the different aspects of the siting and cleanup process, thus highlighting the need for multi-layered analyses to identify and fully understand potential inequities associated with environmental programs.

1. Introduction

In the fall of 1982, the North Carolina state government disposed of 6000 truckloads of PCB laced toxic soil in Warren County (NC DNCR, 2013). This poor, rural county was home to predominately Black populations. Residents from Warren County and surrounding communities took to the streets to physically block the unwelcomed dump trucks. Yet, they were unsuccessful. Despite the unfavorable outcome, this and similar events laid the path for the environmental justice (EJ) movement, including the 1987 seminal study by the United Church of Christ Commission for Racial Justice (UCC, 1987), which was one of the first reports to clearly document and quantify environmental disparities in the siting of toxic waste sites.

This study contributes to the EJ literature by examining whether various “layers” of regulated activities under the Resource Conservation and Recovery Act (RCRA) are disproportionately associated with communities consisting of greater populations of racial and ethnic minorities, and lower income populations. RCRA was first passed in 1976, and gave the U.S. Environmental Protection Agency (EPA) the authority to regulate hazardous material from “cradle-to-grave,” including the

generation, storage, transportation, and eventual disposal or neutralization of the material. Unfortunately, unintended releases of these hazardous chemicals do occur, often due to the degradation of equipment and structures over time, human error, and dated historical practices and land uses. RCRA was amended in 1984 to expand the cleanup provisions and give the EPA the authority to require the investigation and (if needed) remediation of potential releases. Collectively these activities are referred to as a Corrective Action (CA). Approximately 113 million Americans live within three miles of a RCRA CA site (US EPA, 2020).

In this study focus is drawn to the 41,361 RCRA sites in the contiguous US that are Large Quantity Generators (LQGs) of hazardous material; and/or Treatment, Storage, and Disposal Facilities (TSDFs).¹ These types of facilities deal with the largest quantity of chemicals and pose the greatest risks. Among these facilities, 199 CAs involving actual contamination and subsequent remedial actions were initiated within our 2000 to 2019 study period. Remedial actions can include the physical removal and/or neutralization of pollution (e.g., excavate contaminated soil, pump-and-treat groundwater); the construction of physical controls (e.g., build concrete caps over contaminated soil or

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¹ A large quantity generator (LQG) is defined as a facility that generates 1000 kg or more of RCRA-regulated hazardous material in a single month, accumulates 1 kg or more of an acute hazardous material in a single month, or accumulates more than 100 kg of spill material that contains an acute hazardous material (US EPA, 2021).

physical barriers to minimize exposure/migration); and/or institutional controls (e.g., re-zone the land as “industrial” to minimize human exposure). Cleanup solutions— or a combination of solutions— are deemed appropriate if they protect human health and the environment, attain cleanup objectives, and control the source of pollution (US EPA, 2022).

Using nationwide data across the contiguous US, we estimate a series of statistical regression models to answer four primary research questions: (i) Do RCRA-regulated hazardous sites tend to be located in neighborhoods with a higher percentage of Nonwhite, Hispanic, or lower income individuals? (ii) Is a release of contamination and a subsequent CA more likely to occur in such neighborhoods? (iii) Is the duration of investigation and cleanup activities systematically longer in certain communities? And (iv) Is the implementation of less permanent remedial solutions associated with neighborhoods consisting of higher proportions of Nonwhite, Hispanic, and/or lower income populations? Our statistical analyses are comprehensive and intend to identify potential overarching patterns of inequity. At the same time, such a broad-brush analysis can miss out on important site-specific details. As such, this study serves as a useful complement to more detailed case-specific studies (e.g., [Baden and Coursey, 2002](#); [Balayannis, 2020](#); [Boer et al., 1997](#); [Palmer, 2003](#)).

The primary contribution of our study is in investigating multiple facets of an understudied environmental cleanup program using a consistent dataset, study area, and time period. We find evidence that RCRA sites tend to be located in neighborhoods consisting of higher proportions of Nonwhite and Hispanic populations. Communities with higher proportions of Nonwhite and Hispanic populations also face significantly longer release investigation and cleanup times. We find mixed evidence of disproportionate impacts to lower income populations in terms of the likelihood of a release and subsequent cleanup activities. Lastly, there is little evidence of systematic injustices in terms of cleanup permanence; although this result is limited by a fairly small sample size. Overall, the occurrence of disproportionate impacts varies across the different aspects of the siting and cleanup process, thus emphasizing the need for multi-layered analyses to identify and fully understand potential inequities associated with far-reaching environmental programs.

The next section reviews the relevant literature and highlights the specific contributions of this study. Section 3 provides information regarding the source and content of the data that are used. The models and methods used for analysis can be found in section 4, followed by the regression results in section 5. The paper concludes with section 6, where we summarize the primary results, and discuss the EJ implications and limitations of the study.

2. Literature review

Most of the existing literature does not pertain specifically to RCRA-regulated hazardous waste sites, but instead focuses on other environmental programs dealing with hazardous chemicals, including Superfund and the Toxic Release Inventory (TRI). Studies that do examine RCRA often focus on only a subset of sites. Furthermore, much of the literature is local in scope, and the results of such case studies cannot necessarily be extrapolated to the broader universe of sites regulated under national programs. Our current study fills these gaps by empirically assessing multiple facets of the RCRA program, and by doing so at a national level.

Several studies have found injustices with regards to the location of hazardous sites. For example, [Boer et al. \(1997\)](#) found Black and Latino communities in Los Angeles County were more likely to host a TSDF. [Baden and Coursey \(2002\)](#) analyzed 1990 census data and found Superfund sites in Chicago tend to be located in communities with a low population density and higher proportions of Hispanic residents. Superfund sites are industrial facilities that are cleaned up through the Comprehensive Environmental Response, Compensation, and Liability

Act (CERCLA), and often entail the most severe types of contaminated sites. In contrast to their findings based on 1990 demographic data, when examining historical trends in the 1960s, [Baden and Coursey \(2002\)](#) found that hazardous sites were more concentrated in poorer white neighborhoods. The dynamic nature of the socioeconomic factors correlated with hazardous site locations highlights the need to periodically revisit the issue.

It is important to emphasize that facilities handling hazardous materials are not necessarily placed in neighborhoods associated with these characteristics. It could also be that certain populations migrate to neighborhoods with these disamenities due to differences in preferences and income ([Banzhaf and McCormick, 2007](#)). [Been and Gupta \(1997\)](#) attempted to disentangle these effects. They found that the percentage of Black and Hispanic individuals in a census tract were significant predictors of whether a tract hosts a TSDF. They suggest that TSDFs tended to be placed in tracts where there was a higher proportion of Hispanics, whereas the proportion of Black individuals tended to increase in tracts where a TSDF was already sited.

[Banzhaf et al. \(2019\)](#) argue that the problem is more complex than a simple what came first, the chicken or the egg type of question. Shifts in demographics that are correlated with changes in pollution could be obscured for several reasons. Initial household sorting due to pollution could have a domino effect of negative neighborhood characteristics that could reinforce the initial sorting patterns. For example, high pollution could lead to decreases in property values and drive out high income earners. This could then leave the community with fewer resources to ensure attractive retailers, public safety, and good school quality. Additionally, an area’s reputation as “dirty” and “unsafe” due to pollution could lead to public stigma, which can remain even after the pollution issue is resolved ([Messer et al., 2006](#)).

[Banzhaf et al. \(2019\)](#) outline four mechanisms that can lead to disamenities being disproportionately located in different neighborhoods: (1) disproportionate siting on the firm side, (2) “coming to the nuisance” on the household side ([Banzhaf and McCormick, 2007](#); [Been, 1994](#); [1997](#); [Hamilton, 1995](#)), (3) market-like coordination of the two, and (4) discriminatory policies and enforcement. [Banzhaf et al.](#) find the fourth mechanism to be unlikely and discuss how activists and non-economists alike agree that the observed injustices are generally not rooted in the desire to harm certain groups. However, they do find that firms tend to locate in communities with higher minority populations. Such firm-side inclinations could be driven by economic incentives. A lack of coordination and protest among populations in some communities may imply lower costs to locate there ([Hamilton, 1995](#)). Land may also be cheaper in such communities, and there could be increased access to labor pools and transportation infrastructure.

Other studies have found negligible evidence of inequitable impacts associated with the siting of hazardous sites. [Anderton et al. \(1994\)](#) did not find evidence that the percentage of minorities (Black, Hispanic, and foreign-born individuals) was a statistically significant predictor of the location of TSDFs. [Davidson and Anderton \(2000\)](#) expanded on that study to not just examine TSDFs, but to include all RCRA-regulated sites in urban areas. They again found no correlation between race and ethnicity with regards to the siting of RCRA facilities. Similarly, [Been and Gupta \(1997\)](#) found that communities with higher poverty levels were no more or less likely to host a TSDF.

[Hamilton \(1993\)](#) found that communities with greater political engagement (measured by voter turnout) are less likely to be chosen for the expansion of existing commercial waste processing facilities. Applying the Coase theorem ([Coase, 1960](#)), [Hamilton](#) explained that facilities in communities with low political engagement are more attractive for expansion because they pose lower costs and risk of community backlash. [Hamilton \(1995\)](#) expanded the study and found that the zip codes targeted for waste processing expansion had 7% more Nonwhite residents. [Hamilton \(1995\)](#) maintained that political engagement was a key explanation for this difference, stating that, “under the current regulatory regime where communities differ in the

degree they may engage in collective action to voice their environmental compensation demands, firms such as hazardous waste facilities will attempt to locate where, *ceteris paribus*, opposition is the least.” (page 129).

Palmer (2003) investigated whether reductions in the use of toxic chemicals at manufacturing firms in Massachusetts were greater across communities with different sociodemographic characteristics. He found that manufacturers in low-income and minority communities tended to adopt fewer techniques to reduce their use of toxic chemicals.

Another branch of the literature has examined emissions of toxic substances across neighborhoods with different demographics. Focusing on air emissions reported to the TRI, Arora et al. (1998) found that the racial composition of the surrounding community was an important predictor of the occurrence of a toxic release, but only in the Southern US and not in other regions.² Elliot et al. (2004) investigated the relationship between community sociodemographics and accidental acute toxic air emissions at facilities regulated under the EPA's Risk Management Plan (RMP) program. They found that even after controlling for key facility characteristics, facilities located in counties with a high percentage of Black residents and greater income inequality are more likely to experience a chemical accident. In a separate study using the same dataset, Kleindorfer et al. (2003) found that intrinsically hazardous facilities (measured by the quantity of chemicals in inventory and their potential harm) were more likely to be located in minority and impoverished communities. In a more recent look, Guignet et al. (2023) found that these inequities persist, with RMP-regulated chemical facilities tending to still be in close proximity to neighborhoods where a greater proportion of the population is Black, Hispanic, and living in poverty. Furthermore, Guignet et al. found that conditional on the presence of a chemical facility, the occurrence of chemical accidents and subsequent air emissions are more likely to occur as the percentage of the population that is Black, Hispanic, and living in poverty increases.

A few studies also examined the distributional implications of cleanup duration. Lavelle and Coyle (1992) found that it took longer for Superfund sites to get approved cleanup plans in zip codes with predominantly minority populations. They found that in more than half of the programs across the 10 autonomous regions authorized to implement EPA programs, cleanup starts 12%–42% later for Superfund sites in minority neighborhoods, compared to predominately White neighborhoods. Lavelle and Coyle (1992) also found inequities in terms of prevention, suggesting that regulation penalties were 500% higher in zip codes with the greatest White population. The implication is that required safe practices may have been more often enforced in predominantly White neighborhoods.

Also focusing on Superfund site cleanups, Solatyavari et al. (2022) found longer cleanup times in census tracts with a greater percentage of minority populations. Political involvement and percent owner-occupied housing were correlated with faster cleanup times. Sigman (2001) also found that communities with higher voter turnout (a proxy for community coordination and engagement) experience faster Superfund cleanups. Burda and Harding (2014) investigated the time between when a site was first listed as a Superfund site and cleanup completion. They found that sites listed at the beginning of the Superfund program (early 1980s) experienced longer cleanup times when located in more dense, Black, and lower educated communities. However, these associations disappeared later in the Superfund program's history.

Finally, there is a small and fairly dated literature examining the implementation of various remedial methods and the permanence of cleanup. Lavelle and Coyle (1992) found that releases in minority

communities are 7% more likely to be capped off/contained, as opposed to a more permanent remedy—i.e., removal and treatment of the chemicals. In contrast, Gupta et al. (1996) examined the cleanup decisions at 110 wood-preserving and PCB contaminated Superfund sites, and found no evidence of less permanent cleanup solutions in areas with sizable minority or impoverished populations.

Our study contributes to the numerous branches of this literature by providing a comprehensive analysis of a national program and the multiple layers where disproportional impacts could occur. The bulk of the EJ literature on the siting of hazardous sites relied on data from the 1990s or earlier. We provide an updated assessment and investigate beyond just the siting of hazardous waste facilities. Although others have focused on the distributional implications of toxic air emissions and releases of hazardous chemicals into the air, we focus on releases of toxic chemicals into other media, namely land and groundwater. To our knowledge, the literature examining cleanup duration and permanence has focused on Superfund sites. We empirically analyze the duration and permanence of cleanup at facilities regulated under RCRA. Although both programs deal with the cleanup of hazardous chemicals, Superfund sites remediated through CERCLA are often non-operating, abandoned facilities. RCRA, on the other hand, targets hazardous chemicals at facilities that are currently in operation, and where the owners and operators are often known.

3. Data

A cross-sectional dataset of all census tracts across the contiguous US was obtained from the United States Census Bureau's 2000 Decennial Population Census, which coincides with the beginning of our 2000–2019 study period. Focusing on demographics at the beginning of the study period, prior to any new Corrective Action (CA) activities, reduces endogeneity concerns due to reverse causality. We are interested, for example, in how the likelihood of a CA varies with socioeconomic characteristics of a neighborhood. At the same time, such demographics could later shift in response to CA activities. There is a significant literature examining such demographic shifts in response to changes in environmental quality (e.g., Bakkensen and Ma, 2020; Banzhaf et al., 2019; Been, 1994; Cameron and McConaha, 2006; Hamilton, 1993, 1995).

Data on race, ethnicity, poverty status, and median income were gathered at the census tract level.³ The variables derived for later analysis included population density (measured in 1000s of people per square mile); median annual income (measured in \$1000s, 2021 USD); and percentages of the population that are Nonwhite; of Hispanic, Latino, or Spanish descent; and living below the poverty line.⁴ The census-tract level dataset of neighborhood characteristics was then spatially linked to data on RCRA sites and CAs across the contiguous US.

The RCRA and CA data were compiled from RCRAInfo, a multi-relational database collected and maintained by the EPA.⁵ This database is organized into eight different “modules”, each containing several data tables. The “Handler” module includes facility-level information like whether the facility operates as a TSDF, generates hazardous

³ Median income was reported in 1999 USD and converted to 2021 USD based on the U.S. Bureau of Labor Statistics' Consumer Price Index (<https://www.bls.gov/cpi/tables/supplemental-files/historical-cpi-u-202204.pdf>, accessed 13 Oct 2022).

⁴ The percent Nonwhite variable was calculated based on the proportion of the population that identified as only a single race, which comprised 97.5% of the total population. The other socioeconomic variables are based on the total population. To calculate population density, we first derived the area of the census tract using a Geographic Information System (GIS) and the census tract polygon borders from the U.S. Census Bureau's 2000 TIGER/Line files (U.S. Census Bureau, <https://www.census.gov/>, accessed 13 Oct 2022).

⁵ EPA, “RCRAInfo Public Extract”, <https://rcrapublic.epa.gov/rcra-public-export/>, data downloaded in June 2019.

² Our discussion here focuses on hazardous air pollutants, but there is a larger environmental justice literature on air pollution more broadly, mainly in the context of particulate matter and other criteria air pollutants (e.g., Ard, 2015; Clark et al., 2017; Currie et al., 2020).

materials, handles specific types of hazardous materials, and industry classification. The “Corrective Action” module offers information on the timing and severity of CAs, including the start date and, if applicable, the type of remediation implemented and the completion date. For purposes of this analysis we identify CAs as only those where the investigation revealed actual contamination that required clean up or active efforts to minimize exposure.⁶

The data we gathered from RCRAInfo contained 129,670 RCRA sites in the United States as of 2019. Among these sites, 1610 had a CA with confirmed contamination requiring cleanup.⁷ After limiting the data to the contiguous US and CAs that were initiated during our 2000–2019 study period, we are left with 127,846 RCRA sites; 234 of which had a CA initiated. Finally, we draw focus to the largest and riskiest types of RCRA sites – LQGs and TSDFs. Sites that were only reported as small quantity generators, recyclers, transporters, etc. are excluded. The final sample of RCRA facilities analyzed is $n = 41,668$ RCRA sites, among which 199 had an active CA requiring remediation.

Each of our research questions require a slightly different dataset, based on the merged census tract and RCRA facility data. Following the theme of “layers of injustice”, as the research questions progress attention is drawn to a narrower focus and smaller subsets of the data. To examine if the siting of these RCRA facilities is correlated with neighborhood demographics (Q1), we analyze a cross-sectional, tract-level dataset of all census tracts in the contiguous US. Analysis of whether CAs are systematically more or less likely to occur in different types of neighborhoods (Q2) requires a cross-sectional, facility-level dataset of all LQGs and TSDFs in the contiguous US. Duration analysis of whether time to cleanup completion varies with respect to neighborhood characteristics (Q3) is conducted using a subset of the facility-level data, where only LQGs and TSDFs with an ongoing or completed CA are included. Lastly, focus is further drawn to only LQGs and TSDFs with a completed CA, where we then examine whether the permanence of cleanup is correlated with neighborhood socioeconomic characteristics (Q4).

3.1. Tract-level dataset to examine siting of RCRA sites (Q1)

There are 63,083 census tracts in the contiguous U.S. (based on the 2000 Census tract definitions).⁸ As shown in Table 1, 30.5% of tracts contained at least one LQG or TSDF, and the average tract contains 0.52 facilities. In 2000, the average tract population consisted of 23.6% Nonwhite and 11.2% Hispanic. About 13.3% of the population in a tract was living below the poverty line, and the median income was \$71,788, on average. The average population density was 4343 people per square mile.

⁶ These CAs are identified within RCRAInfo based on the following event codes: CA550RC (remedy construction); CA770GW and CA770NG (groundwater and non-groundwater controls); and CA772EP, CA772GC, CA772ID, and CA772PR (institutional controls). See Guignet and Nolte (2024) for further details.

⁷ Our dataset excludes a small subset of facilities that deal with international imports and exports of hazardous materials, as well as facilities that fall under the conditional subpart K exemptions, which apply to academic institutions, such as teaching hospitals and universities (US EPA, 2008).

⁸ We exclude outliers where the census tract had more than seven LQGs or TSDFs (536 tracts), the population density was greater than 60,000 people per square mile or less than two people per square mile (637 and 636 tracts, respectively). These cutoffs roughly correspond to the top or bottom percentiles of the sample. Two additional tracts are excluded because the total population self-identifying as a single race was zero, and that value is used as the denominator in calculating our percent Nonwhite variable. Our estimating dataset of 63,083 tracts maintains 97% of the total number of census 2000 tracts in the contiguous U.S.

Table 1
Descriptive statistics of all year 2000 census tracts.

Variable	Mean	Std. dev.	Min	Max
RCRA site	0.3053	0.4605	0	1
# of RCRA sites	0.52	1.01	0	7
% Nonwhite	23.59	26.63	0	100
% Hispanic	11.22	18.59	0	100
% Below poverty	13.29	11.54	0	100
Median income (\$1000 USD)	71.788	33.876	0	326
Population density (1000s/sq mi)	4.343	7.231	0.002	59.997

Note: Descriptive statistics for 63,083 census tracts in the contiguous U.S. from the 2000 decadal census.

3.2. RCRA facility-level dataset to analyze risk of a Corrective Action (Q2)

A total of 41,361 RCRA facilities that are a LQG and/or TSDF are analyzed. A CA is undertaken during our study period at 199 of these facilities (see Fig. 1). As shown in Table 2, the population statistics corresponding to the tract where a RCRA site is located, compared to those in Table 1 for all tracts across the contiguous U.S., suggest greater proportions of the population that are nonwhite, Hispanic, and living in poverty, as well as slightly higher median incomes. These statistics are not directly comparable, however, because the unit of observation here is a RCRA facility. We more formally test for such differences in our later statistical regression models.

Comparison of the RCRA facility characteristics in Table 2 suggests that 97% of the sample of RCRA sites are a LQG, and about 4% are a TSDF. We also see that among the RCRA sites in our sample, some are also small quantity generators (5%), recyclers of hazardous material (1%), and/or accept hazardous waste from other offsite locations (0.7%). Note that these classifications are not mutually exclusive. Finally, among the facilities where the corresponding North American Industry Classification System (NAICS) codes were available, we see that manufacturing, trade and transportation, and services industries are the most common.

3.3. Dataset to model duration of Corrective Action (Q3)

The duration of a CA is calculated as the amount of time from when an investigation and/or cleanup activities were first initiated, until the contamination issues were sufficiently resolved and the case closed.⁹ The dataset used to model the duration of investigation and cleanup activities focuses on 196 (of the 199) RCRA facilities where a CA involving contamination occurred. Two sites were dropped because they had a negative duration listed, likely due to clerical errors.¹⁰ A third site had the same CA open and completion date, but given the remedial technologies implemented, those dates likely do not reflect the true CA duration.

The CA investigation and cleanup was completed during our study period at 77 of the 196 CA sites. Table 3 shows that the average duration of the 77 completed CAs was 7.07 years, but there is noticeable heterogeneity across cases, ranging from two weeks to over 17 years. For the remaining CAs, the investigation and cleanup activities were not yet

⁹ We define the opening date for a CA as the earliest of the following three events: when regulators were first notified of contamination (CA060), the date an investigation was imposed (CA100), or when the initial assessment was complete (CA200). A CA is considered complete based on the earliest recorded date for one of the following events, and only when applicable to the entire site – when the CA was officially terminated (CA999, CA999NF, or CA999RM), or when the required performance standards were achieved (CA900CR or CA900NC).

¹⁰ Although the start dates were listed as January 2004 and March 2000 for two of the CA sites, the CA completion dates for both were entered as January 1, 2000.

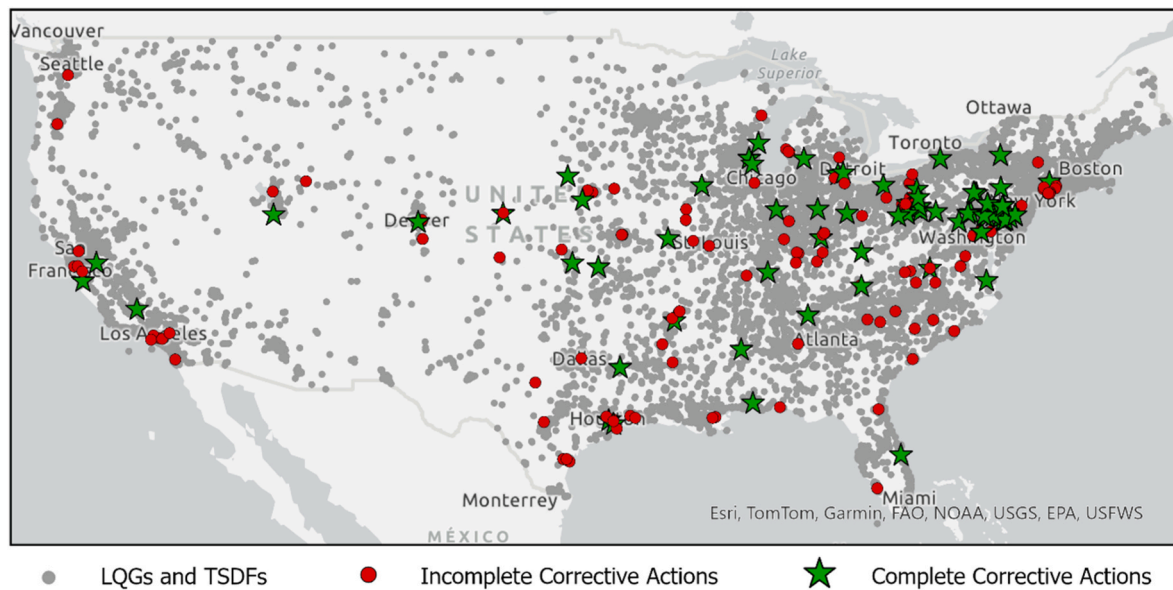


Fig. 1. Large quantity generators (LQGs) and treatment, storage, and disposal facilities (TSDFs) in contiguous U.S.

Table 2
Descriptive statistics of RCRA facilities.

Variable	Mean	Std. dev.	Min	Max
CA Site	0.005	0.069	0	1
% Nonwhite	28.66	26.00	0	100
% Hispanic	16.92	22.32	0	100
% Below poverty	14.08	12.15	0	100
Median income (\$1000s USD)	74.07	34.49	0	326
Population density (1000s/sq mi)	8.512	20.134	0.000	238.627
Large Quantity Generator (LQG)	0.97	0.16	0	1
Small Quantity Generator	0.05	0.22	0	1
Recycler	0.01	0.12	0	1
Treatment, Storage, Disposal Facility (TSDF)	0.04	0.20	0	1
Offsite Waste	0.007	0.085	0	1
Agriculture Ind.	0.013	0.112	0	1
Mining Ind.	0.011	0.104	0	1
Utility Ind.	0.133	0.340	0	1
Construction Ind.	0.046	0.209	0	1
Manufacturing Ind.	0.337	0.473	0	1
Trade and Transportation Ind.	0.345	0.475	0	1
Service Ind.	0.245	0.430	0	1
Waste Management Ind.	0.043	0.204	0	1
Industry Missing	0.036	0.186	0	1

Note: Descriptive statistics for the 41,361 RCRA sites in the contiguous U.S. that are designated as a LQG or TSDF.

Table 3
Additional descriptive statistics of Corrective Actions (CAs).

Variable	Obs	Mean	Std. dev.	Min	Max
Incomplete CAs					
CA Duration (years)	119	14.42	4.23	0.88	19.44
Completed CAs					
CA Duration (years)	77	7.07	5.08	0.04	17.32
Remedy Construction	77	0.88	0.32	0	1
Physical/engineering controls	77	0.25	0.43	0	1
Institutional controls	77	0.57	0.50	0	1
Remedy construction and/or physical/engineering controls	77	0.90	0.31	0	1
Institutional controls only	77	0.10	0.31	0	1

Note: Descriptive statistics for the 199 Corrective Actions that had a start date in or after the year 2000. This includes both complete and incomplete (censored) CAs (77 and 119 CAs, respectively).

complete, as of the time our data was acquired (June 5, 2019). These censored durations, however, still yield useful information and are maintained when estimating our CA duration models (see section 4.3). Among those 119 incomplete CA cases, the average censored duration was 14.42 years.

3.4. Dataset to analyze permanence of cleanup (Q4)

Focusing on the 77 completed CAs, we examine what factors are significantly associated with the permanence of cleanup. Permanence is approximated based on the type of remedy (or remedies) that was implemented at each CA site. We judge “remedy construction” to be the most permanent solution. This entails the construction and implementation of active remedial technologies (e.g., excavation of contaminated soil, pump-and-treat of groundwater). As shown in Table 3, active remediation techniques were used in 88% of the completed CA cases. The remediation types are not mutually exclusive, and may be used in conjunction, but in 25% of the cases we see less permanent physical and engineering controls being implemented to minimize exposure and the migration of contaminants. Such controls include routine monitoring and testing of groundwater, placing concrete caps over contaminated soil, and building other physical barriers. Although cost-effective and perhaps sufficiently effective in many cases, we judge institutional controls (e.g., zoning requirements and restrictions on future land uses) to be the least permanent solution. Among the 77 completed CAs, 57% use institutional controls.¹¹

For our empirical analysis, we aggregate these measures into two indicator variables. The first denotes cases where the most permanent solution – “remedy construction” – is put in place. Other solutions may or may not be implemented in conjunction with these active contamination removal and treatment techniques. The second indicator is more comprehensive, reflecting cases where “remedy construction” and/or physical/engineering controls are put in place. The percentage of completed CA cases corresponding to each are 88% and 90%, respectively. The omitted category, and least permanent solution, entails the 10% of cases where *only* institutional controls are put in place. See Table 3 for details.

¹¹ Similar judgments regarding the relative permanence of different remediation approaches have been made in previous studies (e.g., Gupta et al., 1996; Lavelle and Coyle, 1992).

4. Methods

We next outline the empirical models that are estimated to examine each of our four research questions. The binary models used to analyze key neighborhood sociodemographic factors that are correlated with the siting of a RCRA site (Q1), the probability of a CA occurring (Q2), and whether a relatively more permanent cleanup method is implemented (Q4), are estimated as three sets of Probit models. Models of the duration of the CA at individual facilities (Q4) are estimated as a series of Cox proportional hazard models (Cox, 1972).

4.1. Probit model of the siting of RCRA sites (Q1)

To answer our first research question, we estimate the probability that there is one or more RCRA sites within census tract c as:

$$Pr(site_c > 0) = F(\beta_0 + \mathbf{x}_c\beta_1 + \varepsilon_c) \quad (1)$$

where $site_c$ is the number of RCRA sites within census tract c , and $\mathbf{x}_c$ is a vector of neighborhood characteristics based on the 2000 decadal census. The term ε_c is an assumed normally distributed disturbance term, and $F(\bullet)$ is a normal cumulative density function. The coefficients to be estimated are β_0 and β_1 . To ease interpretation, we present the average marginal effects in our main results tables.

4.2. Probit model of the occurrence of a Corrective Action (Q2)

Although equation (1) is estimated at the neighborhood or tract-level, the remaining models are estimated at the RCRA site-level. The probability of a CA occurring at RCRA site i in census tract c is modelled as:

$$Pr(CA_{ic} = 1) = F(\alpha_0 + \mathbf{x}_c\alpha_1 + \mathbf{z}_i\alpha_2 + \varepsilon_{ic}) \quad (2)$$

where CA_{ic} is a binary dependent variable that equals one if a CA occurs at site i in census tract c during our study period, and zero otherwise. The notation is similar to before. For example, $\mathbf{x}_c$ denotes the sociodemographic characteristics for the census tract where RCRA site i is located. If a tract contains more than one RCRA site, then each site is represented as a separate observation when estimating equation (2). The vector $\mathbf{z}_i$ represents characteristics of the facility, including indicators of whether the facility is a LQG, TSDF, and accepts offsite waste for disposal and treatment. In some models, indicators denoting the type of industry based on aggregated NAICS codes are also included. The vector $\mathbf{z}_i$ is included in some models to assess if CAs may be systematically more or less likely to occur in different types of neighborhoods, even after controlling for other facility-specific risk factors. Similar to the above, we present the average marginal effects when discussing the results of Probit models following equation (2).

4.3. Hazard model of duration of Corrective Action (Q3)

Another layer of potential inequities could occur in terms of the duration of investigation and cleanup activities. In other words, all else constant, do CAs take longer to complete in neighborhoods with greater populations of minorities or those living in poverty? Cox proportional hazard models (Cox, 1972) are often used when examining how covariates impact the duration of the “spell” of interest, in this case of the CA investigation and cleanup. We estimate the hazard or probability that the CA for site i will be completed in time t as:

$$\lambda(t|\mathbf{x}_c, \mathbf{z}_i, risk_i) = \lambda_0(t) \exp(\mathbf{x}_c\gamma_1 + \mathbf{z}_i\gamma_2 + \gamma_3 risk_i) \quad (3)$$

The probability of a CA being completed in a given year is estimated as a function of characteristics of the surrounding neighborhood ($\mathbf{x}_c$) and of the facility itself ($\mathbf{z}_i$). In later models we also include a coarse measure of contamination severity ($risk_i$), which is a 1 to 3 ranking of the EPA’s risk-based cleanup prioritization – 1 = lowest risk and 3 = highest risk.

The coefficients to be estimated are γ_1 , γ_2 , and γ_3 . To facilitate interpretation, we present the corresponding hazard ratios in the main results. For example, the vector of hazard ratios corresponding to marginal changes in neighborhood characteristics $\mathbf{x}_c$, can be expressed as $\exp(\gamma_1)$. The magnitude of the hazard ratio has an intuitive interpretation. A marginal increase in the corresponding variable in $\mathbf{x}_c$, for example, suggests a $(\exp(\gamma_1) - 1) \times 100$ percentage point change in the probability of CA completion in a given year.¹² When the hazard ratio is greater than one, it indicates an increase in the “hazard”, or in our case, an increase in the probability of CA completion in a given year – i.e., a decrease in the CA duration. In contrast, a hazard ratio less than one indicates a lower probability of CA completion in time t , implying an increase in the duration of the CA. As posited by Burda and Harding (2014), the overall premise is that cleanup is occurring equitably if, conditional on facility characteristics and pollution severity, neighborhood demographics are not statistically significant drivers of cleanup duration (i.e., $\exp(\gamma_1) = 1$).

4.4. Probit model of more permanent cleanup approaches (Q4)

To answer our last research question, we estimate a Probit model of the probability that relatively more permanent remediation approaches are used at a CA site. We estimate this model using the subset of RCRA facilities where a CA occurred and was completed during our study period. Let $perm_{ic} = 1$ if a relatively more permanent cleanup approach was utilized, and zero otherwise. More formally:

$$Pr(perm_{ic} = 1) = F(\varphi_0 + \mathbf{x}_c\varphi_1 + \mathbf{z}_i\varphi_2 + \varphi_3 risk_i + \varepsilon_{ic}) \quad (4)$$

We measure $perm_{ic}$ using two alternative variables. The first denotes completed CAs where “remedy construction” was specified as part of the cleanup. This includes active cleanup efforts, such as excavation and removal of contaminated soil, pumping and treating polluted groundwater, etc. The second outcome measure is expanded to include not only remedy construction, but also cleanup approaches based on engineering and physical controls. This second measure is not mutually exclusive, as it includes sites where remedy construction **and/or** physical controls were implemented. The omitted category for this second measure is CA sites where only institutional controls (i.e., zoning and land use restrictions) are put in place. The coefficients to be estimated are φ_0 , φ_1 , φ_2 , and φ_3 . Again, similar to the premise of Burda and Harding (2014), the permanence of a cleanup approach could be considered equitable if, conditional on facility characteristics and pollution severity, the neighborhood demographic variables are statistically insignificant (i.e., $\varphi_1 = 0$).

5. Results

The results for each research question and layer of analysis are presented next. The main results are discussed in terms of average marginal effects or risk ratios, depending on the model.

5.1. Probit results of the siting of RCRA sites (Q1)

Following equation (1), Model 1.1 in Table 4 is the simplest model of the probability of at least one RCRA site being in a census tract, as a function of just the percent of the population that is Nonwhite and the percent that is of Hispanic or Latino origin. The results suggest that, on average, a tract with a one-percentage point higher population of Nonwhites is associated with a 0.03 percentage point increase in the probability of a RCRA site being located there. A similar 0.10 percentage point increase in the likelihood of a RCRA site is associated with a one-percentage point increase in the proportion of the population that is

¹² For details see, for example, Wooldridge (2010, Chapter 22) or Solatyavari et al. (2022).

Table 4

Average marginal effects: Probit model of probability of a RCRA site in census tract.

	(1.1)	(1.2)	(1.3)	(1.4)	(1.5)
% Nonwhite	0.000285*** (0.000070)	0.000517*** (0.000073)	0.000637*** (0.000084)	0.000344*** (0.000078)	0.000457*** (0.000088)
% Hispanic	0.001206*** (0.000099)	0.001297*** (0.000099)	0.001341*** (0.000100)	0.001128*** (0.000103)	0.001163*** (0.000104)
Median income (\$1000 USD)		0.000596*** (0.000056)		0.000579*** (0.000056)	
% Below poverty			−0.001530*** (0.000204)		−0.001508*** (0.000204)
Population density (1000s/sq mi)				0.001796*** (0.000283)	0.001894*** (0.000283)
Observations	63,083	63,083	63,083	63,083	63,083
Log-likelihood	−38712.802	−38658.369	−38682.754	−38637.324	−38659.351

Notes: Binary dependent variable denoting whether a RCRA site is in a census tract (=1) or not (=0). Robust standard errors in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.

Hispanic. These average effects are small but statistically significant.

To help put these findings into context, consider the average tract, one with a population that is 23.7% Nonwhite and 11.3% Hispanic. The predicted probability of a LQG or TSDF site being located in this neighborhood is 30.5%. The results in Model 1.1 suggest that if we compare this to a neighborhood with a 50-percentage point greater Nonwhite population (i.e., 73.7% Nonwhite), then the probability of a RCRA site in the community increases 1.4 percentage points. Making this same illustrative 50-percentage point increase for the proportion of the population that is Hispanic, suggests a roughly 6 percentage point increase in the probability of a RCRA site.

In Models 1.2 and 1.3 we add alternative measures of wealth; both suggest a similar finding – that a tract is more likely to host a TSDF or LQG as median income increases, and is less likely to host such facilities

as the proportion of the population living in poverty increases. Although we find no evidence of inequities being experienced by lower income communities with respect to the siting of RCRA sites, the small yet significant disproportional impacts associated with race and ethnicity are robust. The results are similar even when controlling for population density in Models 1.4 and 1.5, which also suggest that RCRA sites are more often located in more urban areas with higher population densities.

5.2. Probit results of the occurrence of a Corrective Action (Q2)

The average marginal effect estimates from a series of Probit models of the occurrence of a CA are presented in Table 5. Model 2.1 is the simplest, estimating the probability of a CA as a function of the percent

Table 5

Average marginal effects: Probit model of probability of a Corrective Action (CA).

	(2.1)	(2.2)	(2.3)	(2.4)	(2.5)	(2.6)	(2.7)
% Nonwhite	−0.000042** (0.000019)	−0.000061*** (0.000019)	−0.000045** (0.000020)	−0.000022 (0.000019)	−0.000010 (0.000020)	−0.000021 (0.000017)	−0.000006 (0.000019)
% Hispanic	−0.000083*** (0.000031)	−0.000086*** (0.000029)	−0.000084*** (0.000031)	−0.000052** (0.000025)	−0.000052** (0.000026)	−0.000045** (0.000022)	−0.000044* (0.000023)
Median Income (\$1000 USD)		−0.000056*** (0.000013)		−0.000049*** (0.000012)		−0.000035*** (0.000011)	
% Below Poverty			0.000012 (0.000034)		0.000024 (0.000033)		−0.000010 (0.000033)
Population Density (1000s/sq mi)				−0.000887*** (0.000186)	−0.000939*** (0.000194)	−0.000628*** (0.000171)	−0.000646*** (0.000177)
LQG						0.001335 (0.001298)	0.001179 (0.001302)
TSDF						0.012203*** (0.001361)	0.012216*** (0.001358)
Offsite Waste						0.002743 (0.001791)	0.002901 (0.001789)
Agriculture Ind.						−0.003554 (0.003137)	−0.003308 (0.003141)
Mining Ind.						−0.001023 (0.002711)	−0.000825 (0.002720)
Utility Ind.						−0.002231 (0.001714)	−0.002375 (0.001714)
Construction Ind.						−0.002158 (0.001661)	−0.002039 (0.001668)
Trade and Transportation Ind.						−0.003895*** (0.000846)	−0.004025*** (0.000848)
Service Ind.						−0.001894** (0.000819)	−0.002046** (0.000821)
Waste Management Ind.						0.003761*** (0.001039)	0.003872*** (0.001038)
Industry Missing						−0.004334** (0.001905)	−0.004287** (0.001904)
Observations	41,361	41,361	41,361	41,361	41,361	41,361	41,361
Log Likelihood	−1241.139	−1229.177	−1241.081	−1196.910	−1205.906	−1039.101	−1043.972

Notes: Binary dependent variable denoting whether a RCRA site has had a CA (=1) or not (=0). Robust standard errors in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.

of the surrounding population that is Nonwhite and Hispanic. The results suggest that, if anything, facilities in tracts with higher proportions of racial and ethnic minorities are less likely to undergo a CA. At face value, this is positive news in terms of EJ. Perhaps contamination incidents are less likely to occur at facilities in these neighborhoods. At the same time, there could be situations where historical contamination exists, but state and federal regulators are less likely to be notified. Further research into the occurrence of contamination releases versus the reporting and initiation of a CA is needed to fully understand the EJ implications. In any case, this result remains somewhat robust across most models in Table 5.

The next two models include alternative variables to control for household wealth. Model 2.2 suggests that a facility is less likely to have a release discovered, and subsequently undergo a CA investigation and cleanup activities, as median income increases. The average marginal effect suggests that a \$1000 increase in median income is associated with a 0.006% decrease in the probability that a CA occurs at that facility. Put another way, facilities in poorer neighborhoods are more likely to have a CA. For example, the probability of a CA at a facility in the “average” neighborhood, where median annual income is \$74,070, is 0.41%.¹³ If the median household income is \$20,000 lower, the model suggests that the probability of a CA increases by 0.11 percentage points. That is a nearly 27% increase in the risk of a CA. Although this is suggestive of inequities in terms of household wealth, this finding is not robust to our alternative wealth measure. When controlling for household wealth based on the percentage of the population living in poverty, the effects are statistically insignificant. This pattern of mixed evidence holds in subsequent models that control for population density and characteristics of the facilities.

Models 2.4 and 2.5 in Table 5 control for population density. The negative and significant estimates suggest that CAs are less likely to occur at facilities in more densely populated areas. All else constant, an increase of 1000 people per square mile is associated with a 0.089 to 0.094 percentage point decrease in the probability of a CA. It is reassuring that riskier facilities tend to be in less populated areas, where fewer people are potentially exposed. Additionally, the results counter beliefs that injustices may be systematically occurring in more urban neighborhoods. At the same time, the results could be interpreted as potential evidence of disproportionate impacts towards more rural communities. The probability of a CA predicted from Model 2.4 for the “average” site, with a population density of 8512 people per square mile, is 0.13%. All else constant, at a more rural site where the population density equals the 25th percentile (456 people per square mile), the predicted risk of a CA increases to 0.65%, a more than 80% increase.

Models 2.6 and 2.7 control for additional characteristics of the RCRA facility, including industry categories. Even after controlling for facility-specific characteristics, the previous findings are generally robust. Facility characteristics are also generally in line with expectations, lending credibility to our results. For example, TSDFs are more likely to experience a CA. When compared to the omitted category of manufacturing, we see that trade and transportation facilities are less likely to experience a CA, on average. In contrast, facilities categorized as “waste management” are more likely to experience a CA.

5.3. Duration of a Corrective Action results (Q3)

We estimate a series of Cox proportional hazard models (Cox, 1972) to examine whether the duration of CA investigation and cleanup activities systematically varies across facilities located in different types of neighborhoods. The estimated hazard ratios are presented in Table 6. As described in section 4.3, a hazard ratio greater than one implies an

increase in the probability that the CA will be completed in a given year, and therefore suggests a shorter duration. In contrast, a hazard ratio less than one implies a decrease in the probability of completion, and thus a longer CA. The statistical tests reported for the estimates in Table 6 correspond to the null hypothesis that the ratio equals one, meaning no statistically significant effect.

In Model 3.1, the significant hazard ratio corresponding to percent Hispanic suggests that a CA is 2.2% ($=0.978-1$) less likely to be complete in a given year if the proportion of the population that is Hispanic is one percentage point greater. This effect becomes more significant and slightly greater as we control for additional neighborhood and facility characteristics, with our most thorough specifications (Models 3.6 and 3.7) suggesting a 4.3% decrease ($=0.957-1$) in the probability of CA completion. The more thorough models in Table 6 also suggest that a one-percentage point increase in the proportion of the population that is Nonwhite is associated with a 2.3%–2.6% decrease in the probability of CA completion in a given year. Together these results suggest possible inequities; the CA investigation and cleanup take longer to complete in neighborhoods where a greater share of the populations are Nonwhite and Hispanic.

There is little evidence of inequities in terms of income and poverty rates in Table 6, but we do find evidence that CA investigation and cleanup activities occur more quickly in more densely populated areas, perhaps reflecting regulators’ priority to protect human health. This is confirmed by the estimated hazard ratios corresponding to risk in Models 3.6 and 3.7, which suggest that sites prioritized for cleanup due to greater risks to the public and environment are remediated much faster. Increasing the 1 to 3 risk rating by one level suggests a roughly 167% increase in the likelihood that a CA will be completed in a given year.

5.4. Probit of more permanent cleanup approaches (Q4)

Using the sample of 77 completed CAs, we estimate two sets of Probit models of the probability that more permanent cleanup solutions are implemented. The first set is of the dependent variable denoting “remedy construction”, which involves active treatment, neutralization, and/or removal of hazardous chemicals. The second set is of whether remedy construction and/or engineering and physical controls were put in place. The coefficient estimates are presented Tables A.1 and A.2 in the Appendix. Across all models and both dependent variables, we find no evidence that neighborhood and facility characteristics are significantly associated with the type of cleanup approaches that are implemented. It is important to note that this result could be partly driven by our small sample of 77 completed CAs.

6. Conclusion

An estimated 12% of the U.S. population (about 35 million people) live within one mile of a RCRA Corrective Action (CA) site (EPA, 2013). Exposure to the hazardous chemicals used at these facilities can lead to increased risks of cancer, birth defects, organ damage, and other adverse health effects (EPA, 2013). Each of the nearly 130,000 RCRA sites located across the U.S. has the potential to cause long-term damage to local populations, if chemicals are mishandled. Focusing on the largest, and often potentially most hazardous types of RCRA sites – Large Quantity Generators (LQGs) and Treatment, Storage, and Disposal Facilities (TSDFs) – we examine whether the potential for exposure is disproportionately experienced by different subgroups of the population based on race, ethnicity, income, and other dimensions.

Our study contributes to the environmental justice (EJ) literature on hazardous chemicals and industrial facilities in several respects. We provide an updated, layered assessment by going beyond just the siting of hazardous sites. We examine (i) whether RCRA-regulated facilities tend to be located in neighborhoods with a higher percentage of Nonwhite, Hispanic, or lower income individuals, (ii) whether chemical

¹³ More specifically, this is the predicted probability for a site in a neighborhood where the proportions of the population that are Nonwhite, Hispanic, and the median income, are evaluated at the sample means (shown in Table 2).

Table 6

Hazard ratios: Proportional hazard model of Corrective Action duration.

	(3.1)	(3.2)	(3.3)	(3.4)	(3.5)	(3.6)	(3.7)
% Nonwhite	0.993002 (0.006268)	0.992281 (0.006104)	0.987756* (0.006231)	0.988026 (0.007695)	0.985554* (0.007350)	0.977176*** (0.008594)	0.973962*** (0.008253)
% Hispanic	0.978057** (0.010741)	0.978052** (0.010587)	0.978200** (0.009947)	0.970894** (0.011784)	0.972009** (0.010975)	0.956719*** (0.016329)	0.957006*** (0.014762)
Median Income (\$1000 USD)		0.997664 (0.005042)		0.997543 (0.005002)		0.997035 (0.004923)	
% Below Poverty			1.021280** (0.010542)		1.015254 (0.010864)		1.019690 (0.012501)
Population Density (1000s/sq mi)				1.163902* (0.093532)	1.143736 (0.096926)	1.236149** (0.113048)	1.213871** (0.113421)
Large Quantity Generator						1.048888 (0.489751)	0.936188 (0.435418)
TSDF						0.491825 (0.220639)	0.463718* (0.207367)
Offsite Waste						2.251719 (1.176223)	2.285972 (1.183674)
Risk						2.674618*** (0.517368)	2.661281*** (0.519486)
Risk Miss						9.427973*** (5.204926)	9.992352*** (5.505037)
Agriculture Ind.						3.480856 (5.059924)	3.495334 (5.139636)
Mining Ind.						0.000000*** (0.000000)	0.000000*** (0.000000)
Utility Ind.						1.497616 (0.887439)	1.473723 (0.861405)
Construction Ind.						1.505640 (1.146235)	1.536954 (1.110627)
Trade and Transportation Ind.						1.081971 (0.388662)	1.143241 (0.410169)
Service Ind.						0.990105 (0.361207)	0.950677 (0.344990)
Waste Management Ind.						1.813130* (0.637535)	1.821287* (0.637192)
Industry Missing						11.918981*** (5.320164)	11.800940*** (5.344547)
Observations	196	196	196	196	196	196	196
Log likelihood	-373.013	-372.889	-371.798	-370.819	-370.287	-347.386	-346.670

Notes: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The p-values here correspond to the null hypothesis that the hazard ratio estimates are equal to one (see section 4.3 for details).

releases and subsequent CAs are more likely to occur in such neighborhoods, (iii) if the duration of investigation and cleanup activities are systematically longer in different types of communities, and (iv) whether the implementation of less permanent cleanup approaches is associated with neighborhoods consisting of higher shares of Nonwhite, Hispanic, and/or lower income populations. Much of the literature has focused on abandoned, often non-operational Superfund sites. In contrast, sites regulated under RCRA are usually active industrial facilities.

We find evidence of disproportionate impacts across racial, ethnic, and income dimensions, but these findings vary for different layers of the RCRA program. There are small, yet statistically significant, positive relationships between greater shares of Nonwhite and Hispanic populations, and the likelihood of a RCRA site being present. Conditional on the presence of a RCRA site, however, we find that the probability of a CA investigation and cleanup is less likely at sites in communities with a greater proportion of minority populations. It could be that releases are less likely to occur in these communities, but this result could also reflect that releases that do occur are less likely to be detected and reported. Further research is needed to disentangle these two effects.

We find mixed evidence as to whether CAs are more likely at facilities in lower income communities. The results suggest no statistically significant association between the percent of the population with an income below the poverty line and the occurrence of a CA, but we do find that a release and subsequent cleanup is more likely at facilities in communities with lower median income levels. For example, compared to a facility in the average neighborhood, a site in a community with a

\$20,000 lower median income is 0.11 percentage points more likely to undergo a CA due to contamination issues. This may seem small, but it represents a 26.9% increase compared to the average baseline risk of 0.41%.

Among facilities where a CA does occur, we find that the investigation and cleanup take significantly longer when located in communities where greater shares of the population are Nonwhite and Hispanic. The probability of a CA investigation and cleanup being completed in a given year decreases by 2.2–4.3 percentage points for each one percentage point increase in the proportion of the local population that is Nonwhite or Hispanic. It could be that these facilities pose more complicated and severe pollution issues, and thus require more thorough (and lengthier) cleanups. However, our findings of inequitable cleanup durations are in agreement with past studies (e.g., [Lavelle and Coyle, 1992](#); [Solatvavari et al., 2022](#); [Burda and Harding, 2014](#)), and remain robust even after controlling for characteristics of the facility and our coarse measure of contamination severity. As such, a more likely explanation may be that certain communities are being neglected or do not have the resources to organize and push regulators and responsible parties for quicker recoveries. Such speculative explanations cannot be fully teased out from the coarse data used in our broad-brush analysis here, but more locally focused research and community engagement could shed light on these issues.

Lastly, we find no robust, statistically significant associations between local socioeconomic characteristics and the permanence of the cleanup activities undertaken. Our null findings suggest that no disproportionate impacts are occurring with respect to the permanence

of cleanup. This is in line with Gupta et al.'s (1996) analysis of Superfund sites, but contrasts with Lavelle and Coyle (1992), who found that less permanent containment-focused strategies are more likely to be implemented in minority communities. The sample size for this part of our analysis is small, consisting of just 77 RCRA sites where a CA was complete; and so caution is warranted when interpreting this result.

There are a few other important caveats to keep in mind when interpreting our results. These shortcomings also highlight important directions for future research. Although our analysis is comprehensive, it provides only a broad-brush, coarse look at the distributional implications of RCRA. Our estimates reflect the average associations across the contiguous U.S. A null average effect does not necessarily imply that there are not inequitable impacts occurring at the regional or even individual community levels in some instances. Due to our reliance on spatially and temporally coarse Census data, our analysis could also be missing important dynamic and local, within-tract effects. Additionally, although we strive to examine several different types of socioeconomic characteristics that partly define different subpopulations, there may be important dimensions that we did not consider, such as linguistic isolation, immigration status, age, etc.

We emphasize that the causality of our estimated relationships is ambiguous. We see the identification of potential disproportionate impacts across aspects of the RCRA program as an important first step, especially at this comprehensive nationwide level. But if the policy goal is to minimize potentially disproportionate effects, then further research is needed to identify the causal mechanisms driving these inequities.

Lastly, we caution any extrapolation of our findings to hazardous chemicals and other environmental programs outside of the U.S., especially towards developing countries that may have very different histories in terms of population dynamics, systemic inequities, and environmental pollution and regulations. Nonetheless, our methodological approach can provide insights to inform research of other environmental programs and in other countries. Researchers and practitioners can apply similar empirical modelling techniques to those other contexts. When doing so, we emphasize the need to think outside of the box and analyze the numerous different layers of a program. As demonstrated in our study and in the literature more broadly, there are many ways in which inequitable impacts can surface.

Overall, we find that the EJ implications surrounding RCRA and the investigation and cleanup activities through the Corrective Action program vary across the different aspects of the siting and cleanup process. This highlights the need for multi-layered analyses when trying to

identify and fully understand potential inequities associated with broad-scale environmental programs. Such perspective is timely given renewed calls to solidify the role of distributional analyses and equity considerations in federal regulatory development (Biden, 2021).

CRediT authorship contribution statement

Sarah Brodin: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dennis Guignet:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dennis Guignet reports financial support was provided by Oak Ridge Institute for Science and Education. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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APPENDIX

Table A1

Average marginal effects: Probit model of more permanent cleanup technique (Construction remedy).

	(4.1A)	(4.2A)	(4.3A)	(4.4A)	(4.5A)	(4.6A)	(4.7A)
% Nonwhite	−0.001222 (0.001520)	−0.000708 (0.001709)	−0.002195 (0.002024)	−0.000446 (0.001637)	−0.001596 (0.002211)	0.000463 (0.001831)	−0.001194 (0.002260)
% Hispanic	0.002047 (0.004100)	0.002263 (0.004130)	0.002513 (0.004219)	0.000539 (0.004704)	0.000884 (0.005286)	−0.000636 (0.005838)	−0.000368 (0.006190)
Median Income (\$1000 USD)		0.000951 (0.001261)		0.001206 (0.001284)		0.001472 (0.001151)	
% Below Poverty			0.002736 (0.003511)		0.001326 (0.004479)		0.002381 (0.004451)
Population Density (1000s/sq mi)				0.030720 (0.028463)	0.026457 (0.033628)	0.042759 (0.036798)	0.036680 (0.041616)
LQG						−0.189789 (0.146665)	−0.190522 (0.150390)
TDSF						−0.080384 (0.109850)	−0.096616 (0.113379)
Offsite Waste						0.023273 (0.112217)	0.031943 (0.115912)

(continued on next page)

Table A1 (continued)

	(4.1A)	(4.2A)	(4.3A)	(4.4A)	(4.5A)	(4.6A)	(4.7A)
Risk						−0.024529 (0.061918)	−0.020349 (0.060527)
Risk Missing						−0.078812 (0.194275)	−0.035057 (0.193009)
Observation	77	77	77	77	77	77	77
Log Likelihood	−27.429	−27.255	−27.246	−26.605	−26.851	−25.454	−25.772

Notes: Binary dependent variable denoting whether a CA site has been remedied using remedy construction (=1), or not (=0). Robust standard errors in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.

Table A2

Average marginal effects: Probit model of more permanent cleanup technique (Construction remedy and/or engineering and physical controls).

	(4.1B)	(4.2B)	(4.3B)	(4.4B)	(4.5B)	(4.6B)	(4.7B)
% Nonwhite	−0.001270 (0.001419)	−0.001114 (0.001584)	−0.003091 (0.001904)	−0.000877 (0.001519)	−0.015503* (0.009417)	−0.000277 (0.001685)	−0.002651 (0.001923)
% Hispanic	0.001204 (0.003691)	0.001274 (0.003678)	0.002007 (0.003924)	−0.000187 (0.004276)	0.021365 (0.024692)	−0.001273 (0.005556)	−0.000421 (0.005888)
Median Income (\$1000 USD)		0.000285 (0.001068)		0.000513 (0.001099)		0.000616 (0.000908)	
% Below Poverty			0.005298 (0.003644)		0.027804 (0.018360)		0.005779 (0.004062)
Population Density (1000s/sq mi)				0.025861 (0.027169)	−0.047824 (0.103956)	0.033149 (0.034684)	0.024043 (0.039321)
Large Quantity Generator						−0.150912 (0.139599)	−0.165616 (0.141323)
TDSF						−0.102683 (0.106102)	−0.123009 (0.105499)
Offsite Waste						0.019178 (0.107239)	0.025422 (0.107361)
Risk						0.015919 (0.055110)	0.022758 (0.050542)
Risk Missing						0.017149 (0.174487)	0.077956 (0.164279)
Observation	77	77	77	77	77	77	77
Log Likelihood	−25.272	−25.254	−24.629	−24.725	−24.413	−23.655	−23.108

Notes: Binary dependent variable denotes whether a CA site has been remedied using remedy construction and/or engineering and physical controls (=1), or not (=0). The latter (=0) corresponds to cases where only institutional controls are put in place. Robust standard errors in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.

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